

SAMBA SERVER

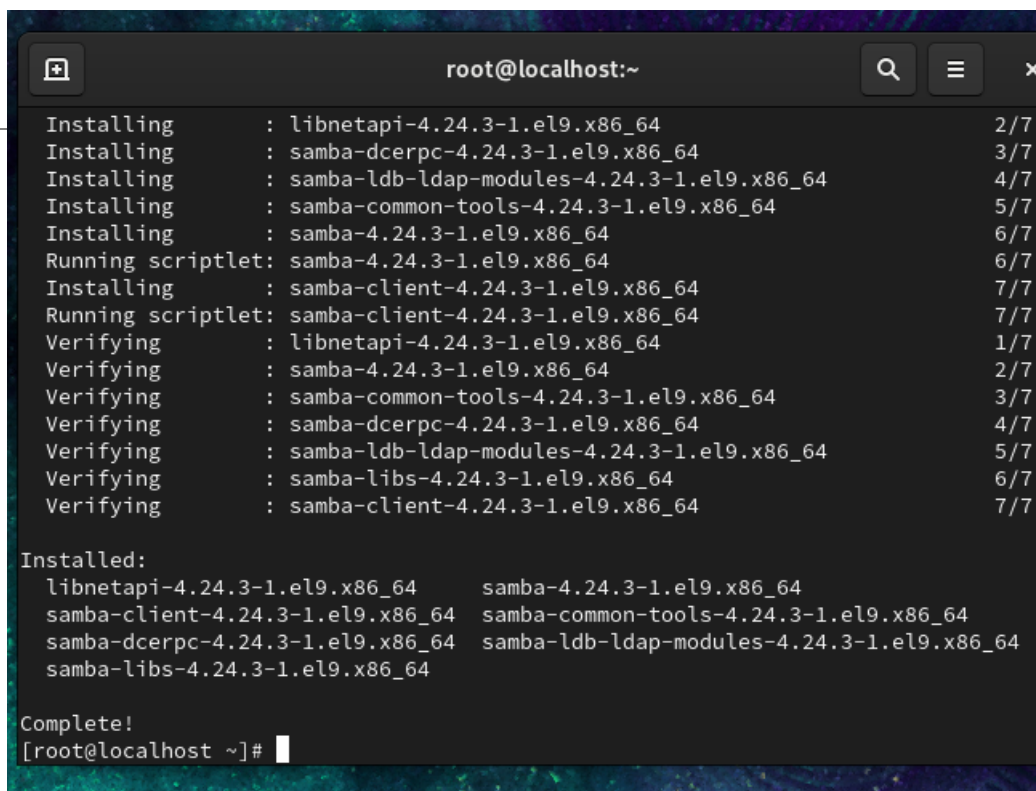
TASKS

1. Install Samba on your Linux system and create a shared folder named 'music-share' that is accessible from Windows machines on your local network.

Install Samba and Create Shared Folder

Step 1: Install Samba

`dnf install samba samba-client -y`



```
root@localhost:~  
Installing      : libnetapi-4.24.3-1.el9.x86_64                2/7  
Installing      : samba-dcerpc-4.24.3-1.el9.x86_64            3/7  
Installing      : samba-ldb-ldap-modules-4.24.3-1.el9.x86_64  4/7  
Installing      : samba-common-tools-4.24.3-1.el9.x86_64      5/7  
Installing      : samba-4.24.3-1.el9.x86_64                  6/7  
Running scriptlet: samba-4.24.3-1.el9.x86_64                  6/7  
Installing      : samba-client-4.24.3-1.el9.x86_64            7/7  
Running scriptlet: samba-client-4.24.3-1.el9.x86_64            7/7  
Verifying       : libnetapi-4.24.3-1.el9.x86_64                1/7  
Verifying       : samba-4.24.3-1.el9.x86_64                  2/7  
Verifying       : samba-common-tools-4.24.3-1.el9.x86_64      3/7  
Verifying       : samba-dcerpc-4.24.3-1.el9.x86_64            4/7  
Verifying       : samba-ldb-ldap-modules-4.24.3-1.el9.x86_64  5/7  
Verifying       : samba-libs-4.24.3-1.el9.x86_64              6/7  
Verifying       : samba-client-4.24.3-1.el9.x86_64            7/7  
  
Installed:  
libnetapi-4.24.3-1.el9.x86_64      samba-4.24.3-1.el9.x86_64  
samba-client-4.24.3-1.el9.x86_64  samba-common-tools-4.24.3-1.el9.x86_64  
samba-dcerpc-4.24.3-1.el9.x86_64  samba-ldb-ldap-modules-4.24.3-1.el9.x86_64  
samba-libs-4.24.3-1.el9.x86_64  
  
Complete!  
[root@localhost ~]#
```

Step 2: Start Samba Services

`systemctl enable smb nmb`

`systemctl start smb`

`systemctl start nmb`

Verify:

`systemctl status smb`

```
[root@localhost ~]# systemctl status smb
● smb.service - Samba SMB Daemon
   Loaded: loaded (/usr/lib/systemd/system/smb.service; enabled; preset: disabled)
   Active: active (running) since Fri 2026-07-17 15:15:10 IST; 27s ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
  Main PID: 19808 (smbd)
    Status: "smbd: ready to serve connections..."
      Tasks: 3 (limit: 4273)
    Memory: 22.8M (peak: 23.0M)
       CPU: 199ms
    CGroup: /system.slice/smb.service
            └─19808 /usr/sbin/smbd --foreground --no-process-group
              └─19810 notifyd
                └─19811 cleanupd

Jul 17 15:15:09 localhost.localdomain systemd[1]: Starting Samba SMB Daemon...
Jul 17 15:15:10 localhost.localdomain systemd[1]: Started Samba SMB Daemon.
lines 1-18/18 (END)
```

Step 3: Create Shared Folder

```
mkdir -p /srv/music-share
chmod 777 /srv/music-share
```

Verify:

```
ls -ld /srv/music-share
```

```
[root@localhost ~]# ls -ld /srv/music-share
drwxrwxrwx. 2 root root 6 Jul 17 15:16 /srv/music-share
[root@localhost ~]#
```

2. Configure Samba to allow only users 'partyuser' and 'guestuser' to access the 'music-share' folder, and set a Samba password for each user.
Hint: Use the `smbpasswd` command to set Samba passwords after creating the Linux users.

Configure Samba Users

Step 1: Create Linux Users

```
useradd partyuser
passwd partyuser
```

Create the second user:

```
useradd guestuser
passwd guestuser
```

Step 2: Add Samba Passwords

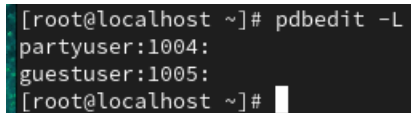
```
smbpasswd -a partyuser
```

Then:

```
smbpasswd -a guestuser
```

Step 3: Verify Samba Users

```
pdbedit -L
```



```
[root@localhost ~]# pdbedit -L
partyuser:1004:
guestuser:1005:
[root@localhost ~]#
```

3. Edit the smb.conf file to make the 'music-share' folder read-only for 'guestuser' but read/write for 'partyuser'.

Configure Samba Share

Step 1: Backup Configuration

```
cp /etc/samba/smb.conf /etc/samba/smb.conf.bak
```

Step 2: Edit Samba Configuration

```
vi /etc/samba/smb.conf
```

Go to the bottom and add:

```
[music-share]
path = /srv/music-share
browseable = yes
valid users = partyuser guestuser
read only = no
write list = partyuser
read list = guestuser
create mask = 0664
directory mask = 0775
```

Save:

Esc

:wq

Step 3: Restart Samba

```
systemctl restart smb
```

```
systemctl restart nmb
```

Step 4: Verify Configuration

testparm

```
[root@localhost ~]# testparm
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_STANDALONE

Press enter to see a dump of your service definitions
```

4. From a Windows machine or VM, connect to the 'music-share' Samba share using the 'partyuser' account and copy a sample MP3 file into the share. Then verify from Linux that the file appears in the shared folder.

Access Share from Windows

Step 1: Open File Explorer on Windows

Press:

Windows + R

Type:

\\192.168.231.128\music-share

Step 2: Login

Username:

partyuser

Password:

Party@123

Step 3: Copy a Sample MP3

touch /srv/music-share/song.mp3

Step 4: Verify from Linux

ls -l /srv/music-share

```
[root@localhost ~]# systemctl restart smb
[root@localhost /]# touch /srv/music-share/song.mp3
[root@localhost /]# ls -l /srv/music-share
total 0
-rw-r--r--. 1 root root 0 Jul 22 17:53 song.mp3
[root@localhost /]#
```

5. Restart the Samba service and use the smbstatus command to check if any clients are currently connected to your Samba share.
Hint: If no clients are connected, try accessing the share from another device first.

Check Samba Connections

Step 1: Restart Samba

```
systemctl restart smb
```

Step 2: Check Connections

```
smbstatus
```

```
[root@localhost /]# systemctl restart smb
[root@localhost /]# smbstatus

Samba version 4.24.3
PID      Username   Group      Machine                                Protocol Version Encryption
Signing
-----
-----

Service   pid      Machine    Connected at                          Encryption Signing
-----

No locked files
[root@localhost /]#
```

Firewall Configuration (Required)

Allow Samba:

```
firewall-cmd --permanent --add-service=samba
```

Reload:

```
firewall-cmd --reload
```

Verify:

```
firewall-cmd --list-services
```

SELinux (If Access Is Denied)

If Windows cannot access the share:

setsebool -P samba_export_all_rw on

Apply the correct SELinux context:

chcon -t samba_share_t /srv/music-share

Verify:

ls -Zd /srv/music-share

```
[root@localhost /]# ls -Zd /srv/music-share
unconfined_u:object_r:samba_share_t:s0 /srv/music-share
[root@localhost /]#
```

